

Anas M. Wagih

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Summary

Senior-year Electronics & Communication Engineering student (graduating 2027) with hands-on experience building and **validating** AI-powered products and end-to-end ML pipelines. Proficient in Python, PyTorch, TensorFlow, and Generative AI APIs (LLMs/GPT), with practical experience testing and verifying **conversational AI systems**, API integrations, and cloud-based training pipelines across open-source projects on [GitHub](#). Passionate about applying ML and GenAI to solve real-world engineering problems at scale.

Education

Arab Academy for Science, Technology, and Maritime Transport — Egypt 09/2022 – 02/2027
 B.Sc. in Electronics and Communication Engineering

- **ABET**-accredited Degree, USA
- **Relevant Coursework:** Machine Learning, Deep Learning, Probability & Statistics, Linear Algebra, Digital Signal Processing, Microprocessors, Embedded Systems, Electrical Circuit Analysis

Experience

Software & IoT Engineering Intern — Clixsys 07/2025 – Ongoing

- Built an **AI-powered in-app assistant** integrating **OpenAI's GPT API** (LLM) for natural-language smart-home control and real-time data retrieval; **validated conversational flows, intent accuracy, and API response correctness across 20+ structured test scenarios**; achieved **≈2 s** end-to-end response latency with multi-voice support.
- Implemented a **Facial Recognition** pipeline (one-shot learning from user-profile images) for automated gate access, achieving **>95% recognition accuracy** and **<500 ms** inference latency on edge hardware.
- Co-developed **3** cross-platform (iOS/Android) applications serving **10k+ active users** using Quasar (frontend) and Firebase/AWS (backend), **reducing average feature delivery time by ~30%** through shared codebase architecture.

Embedded Systems Trainee — National Telecommunication Institute 08/2025 – 09/2025

- Hands-on embedded C programming for AVR microcontrollers across **6 peripheral modules**: GPIO, ADC, timers, interrupts, USART, SPI, and I2C.
- Designed a Dual-MCU STM32-based Deadman Safety and Telemetry System for Electric Vehicles, integrating **2 independent MCUs** with real-time CAN-bus telemetry and sub-**10 ms** fault-response latency.

Projects

BitBlock — **Open-Source No-Code Microcontroller Programming Platform** [[Live Demo](#)]

- Engineered an end-to-end **on-device ML pipeline**: dataset collection → model training (Python / TensorFlow/Keras) → **INT8 post-training quantization** (**~4× model-size reduction**) → TFLite conversion and C header generation for microcontroller deployment, with hardware-gated model compatibility enforced per board RAM/flash constraints.
- Implemented sensor data preprocessing and temporal windowing using **NumPy** and **PIL** (image pipeline), NumPy-accelerated preprocessing pipeline for DSP feature extraction (FFT, windowing, MFCC), optimized for efficient integration with the cloud-based TensorFlow training loop running on Google Cloud Run.
- Architected a browser-based **Blockly** editor that transpiles visual blocks into compiled C/C++ firmware for **10 ESP32/Arduino boards** — zero install, cloud-compiled via containerized toolchains on Google Cloud Run, **reducing firmware build time to under 30s** per compilation.
- Applied **software engineering best practices**: modular OOP design (React/TypeScript frontend, Dockerized backend), automated compilation testing, CI/CD pipeline.

PCBs Portfolio — Open-Source Hardware (10+ Boards) [\[Repo ↗\]](#)

- Designed and published **10+ production-ready PCBs** — custom **ESP32/ESP8266** and **STM32** boards with optimized RF layout and power regulation — under *8bitlab*, a personal open-hardware initiative, **all passing DRC with zero errors**.
- Followed professional practices (DRC, footprint validation, layer stackup); released full schematics, Gerber files, and BOMs for direct fabrication.

Skills

ML / AI: PyTorch, TensorFlow, TensorFlow Lite, Keras, Scikit-learn, NumPy, PIL/Pillow, Pandas, Matplotlib, LLM / GenAI APIs (OpenAI GPT), NLP fundamentals, INT8 quantization, on-device inference, RAG (conceptual).

Programming: Python, Embedded C, JavaScript, MATLAB, C++, SQL.

Tools & Cloud: Git, Postman, AWS, Firebase, Google Cloud Run, Linux, Microchip Studio, STM32CubeIDE, KiCad, Android Studio, Xcode, Arm Mbed.

Hardware: Arduino, Raspberry Pi, ESP32, PCB Design, Power Electronics, Soldering.

Languages: Arabic (Native) & English (Proficient — Cambridge IGCSE O-Level: **“A” grade**)

Courses & Training

- DeepLearning.AI — Deep Learning Specialization (5 Courses) [\[Link ↗\]](#)
- DeepLearning.AI + Stanford University — ML Specialization (3 Courses) [\[Link ↗\]](#)
- Arm Education — Arm Cortex-M Architecture & Software Development (4 Courses) [\[Link ↗\]](#)
- Arm Education — Advanced Embedded Systems on Arm (2 Courses) [\[Link ↗\]](#)

Activities

Machine Learning Tutor — ECE Department ML Workshop (*Linear & Logistic Regression, KNN; 20+ students*) [↗](#)

Arduino Co-Tutor — ECE Department hands-on Arduino Workshop (*25+ students per session*)

University Robotics Club — Electrical Team Leader